

Beacon Data Processing Systems

Ohlone College

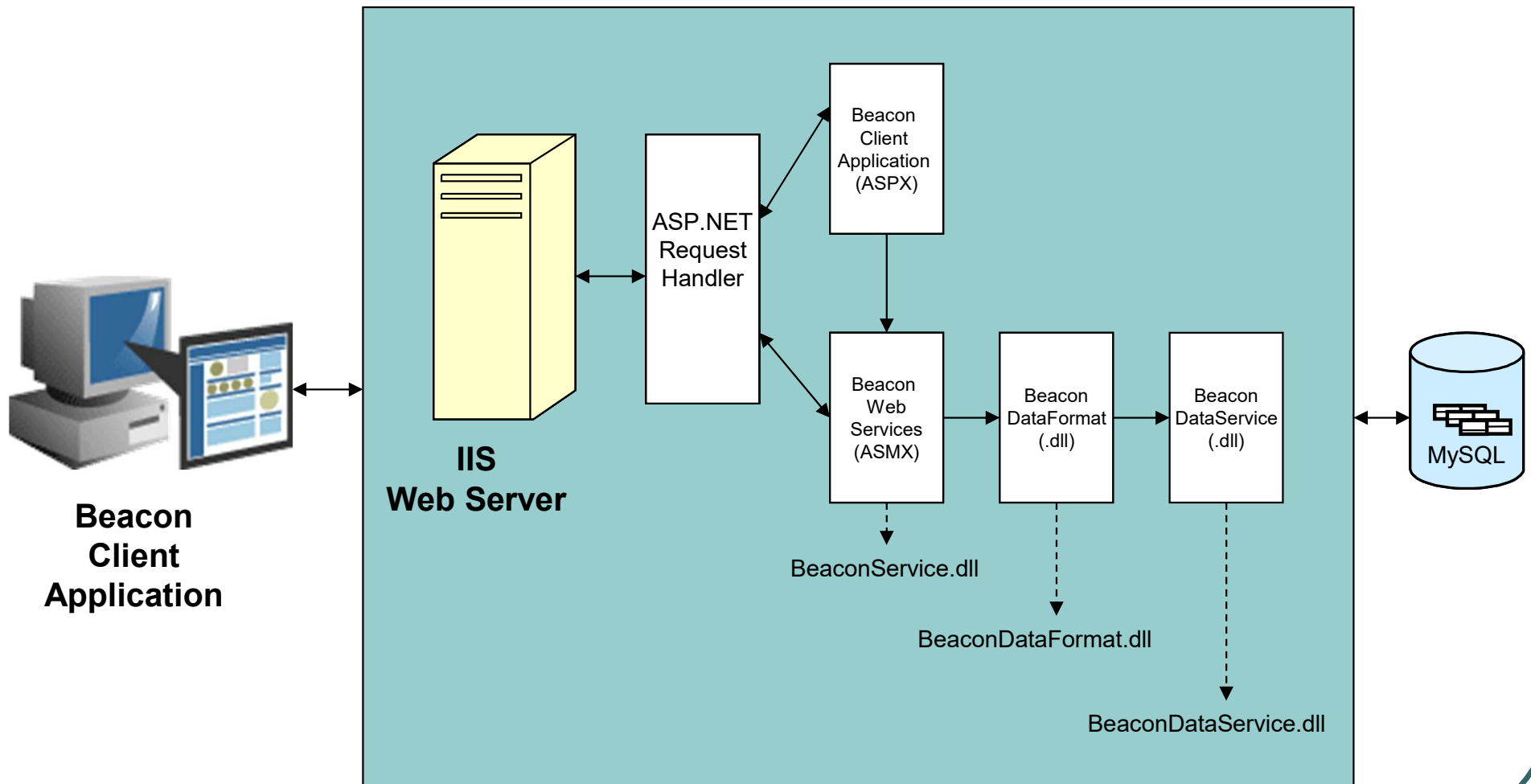
Team Members

Bruce Davis,
Edward Anthonymsamy,
Gopalakrishnan Veeraragavan,
Ryan Edge

Faculty Advisors

Jim Pham
David Topham,

Beacon Data Processing Systems Architecture



Application Modules

1. Bean Client Application (ASPX)
2. Beacon Web Services (ASMX)
3. Beacon Data Format (.DLL)
4. Beacon Data Service (.DLL)
5. MySQL

Bean Client Application Module

- Beacon client is a Web-based application using the .NET ASPX model.
- ASPX is essentially web page that implements the .NET WebForm and C#.

Beacon Web Services Module

- Beacon Web Services (WS) are a set of public APIs that are responsible for the beacon data format, decoding, and performing the business logic.
- Beacon Web Services use an industry-specific messaging standard (SOAP) to be interoperable between the beacon client application and the beacon data format.
- These general public WS can also be used by developers, NASA, schools, etc., to perform the business logic.

Beacon Data Format Module

- This module is the business logic for the beacon data format, such as decoding, converting, calculating, etc.
- Before sending the beacon data to the database, the beacon data is first decoded, converted, and calculated.
- This module works along with the beacon Web services Module and beacon data service modules.

Beacon Data Service Module

- Beacon Data Service responsible for the data access that include the database connectivity as well access the database.
- This module is the data tier of the beacon data processing systems.

Database Tables

Beacon Client Application

Untitled Page - Windows Internet Explorer
http://localhost:55966/Submission.aspx

File Edit View Favorites Tools Help

hp Y! Web Search Bookmarks Settings Messenger Mail Music Wikipedia

Norton Phishing Protection on Identity Safe Log-ins

Untitled Page

NASA Home > Beacon Data Submission

Beacon Data Processing System (BPS)

Home
▶ **Beacon Data Submission**
▶ Data Analysis
▶ Contact Us

Welcome to Beacon Data submission page

Thank you for your interest in our mission. If you have received beacon messages from the PharmaSat spacecraft, please use the form below to submit the messages. You will receive a QSL confirmation card about the contact.

Name: Call Sign:

Email: Location:

[Beacon Packets](#)

http://www.yahoo.com/?fr=fptb-hptb6 Internet | Protected Mode: On 100%